

Bearing Remaining Useful Life Prediction Using an Order-Domain Transformer and Causal End-of-Life Smoothing

회전 차수 영역 기반 트랜스포머와 인과적 종료시점 평활화를 이용한 베어링 잔여수명 예측

Hajun Jang^{1,2,†}, Sanha Jang¹, Andrew H Kim¹, Hansol Lim¹ and Jongseong Brad Choi^{1,2,3,*}

¹Department of Mechanical Engineering, SUNY Korea, Incheon, 21985, Korea

²Department of Mechanical Engineering, Stony Brook University, Stony Brook, NY 11794, USA

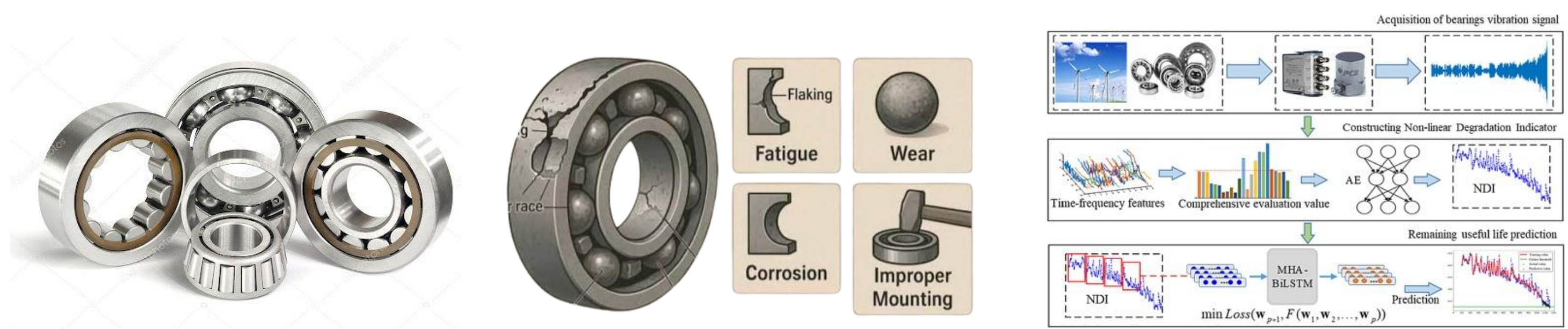
³Department of Computer Science, Stony Brook University, Stony Brook, NY 11794, USA

[†]Corresponding Author

Abstract

Predicting the remaining useful life (RUL) of bearings from vibration signals under varying operating conditions is a challenging task, because speed variations distort defect-related features and time-series predictions often exhibit temporal instability. To address these issues, this study proposes a bearing RUL prediction framework that combines order-domain feature representation with causal time-series post-processing. First, rotational-speed-based order tracking is applied to transform vibration signals from the time domain into the order domain, so that defect related components synchronized with shaft rotation remain consistently aligned despite speed fluctuations. Next, a Transformer-based prediction model is constructed using segment-level order-domain features as inputs, and RUL is estimated in a causal manner while preserving temporal order. In addition, to improve the physical plausibility and temporal stability of the predicted RUL trajectories, a causal end-of-life smoothing method is introduced. This method adjusts predictions using only past information derived from the predicted end-of-life at each time step, thereby preventing future information leakage and making the framework suitable for online applications. The proposed framework provides a practical approach for bearing RUL prediction by jointly achieving speed-robust feature representation and temporally consistent prediction.

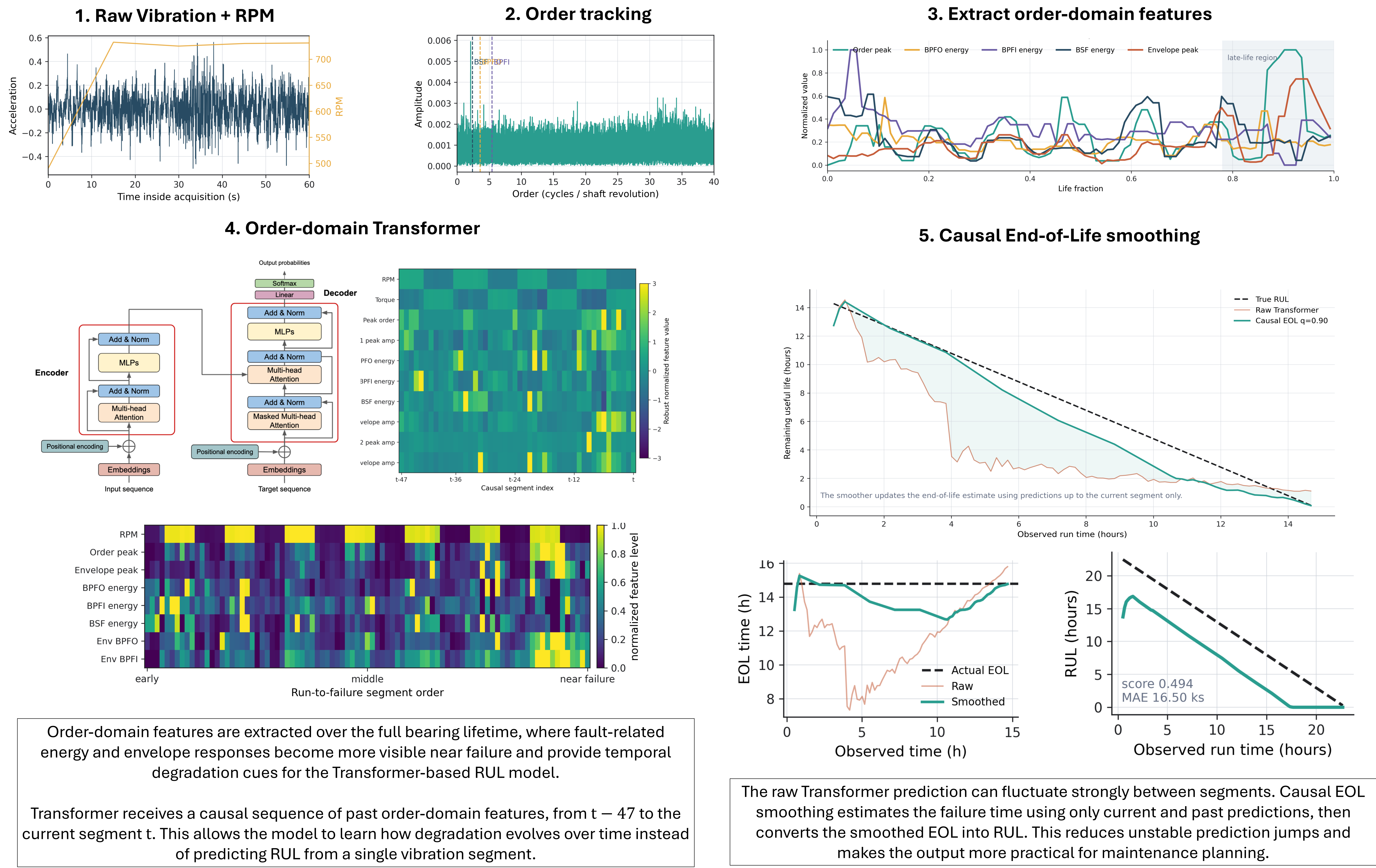
Why predict Bearing RUL?



This method supports predictive maintenance for variable-speed rotating machinery by converting speed-dependent vibration signals into stable order-domain features for earlier and steadier RUL prediction.

- Stabilizes bearing fault patterns under changing RPM
- Reduces noisy or unstable RUL estimates
- Detects degradation trends earlier
- Helps schedule maintenance before failure
- Reduces unexpected downtime and emergency repair cost

Methodology



Results (Remaining Useful Life)

